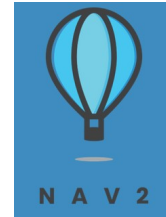


An implementation of Open-RMF in an office environment

Objectives

- At **Elettra Sincrotrone** with the **IT Group**, we wanted to see the Open Robotics Middleware Framework (**Open-RMF**) capacities for use in our warehouse and offices and upgrade our Autonomous Mobile Robots (**AMRs**) to ROS 2, so we:
 - Build 2 AMRs with ROS 2 and Nav2 creating 2 fleets:
 - One from an UGV **Mini**
 - One porting a **Jobot** AMR from ROS 1 to ROS 2
 - Port the **DStar** global planner from ROS 1 to ROS 2 to be used in our robots
 - And, finally, use Open-RMF as middleware for these two fleets

ROS 2

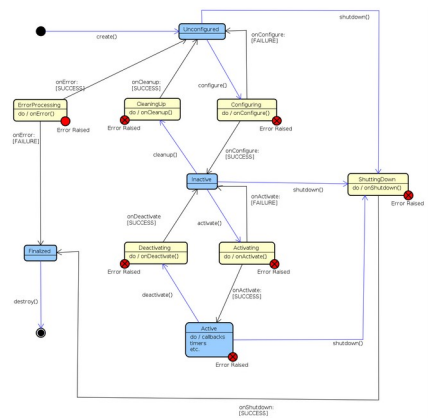


Open-RMF

ROS 2 Nav2 structure

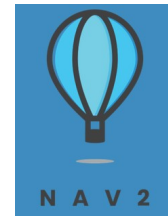
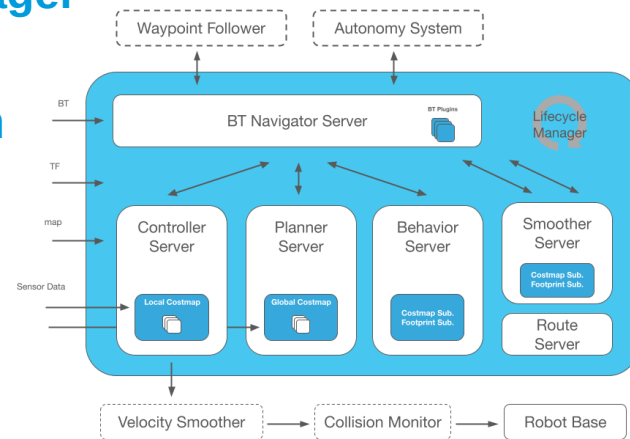
Nav2 structure

- **Behavior Tree Navigation Server**
- **Planner Server**
- **Smoother Server**
- **Controller Server**
- **Behavior Server**
- **Route Server**
- **Lifecycle manager**



Lifecycle

```
require:
base_link → odom
odom → map
```



Upgrade Jobot AMR - ROS 1 → ROS 2

- Build system: catkin → colcon (ament_cmake)
- rclcpp with more C++ 17 features, like smart pointer and new cmake
- Naming
- ROS msgs update
- Lifecycle implemented
 - In on_configure() : add creation of secondary nodes (IMU), setup parameters and create callbacks
 - In on_deactivate() : cleaning
- Find ros-serial for ROS 2

Updated DStar global planner and results

- Upgrade planner
 - Inheritance from `nav2_core::GlobalPlanner`
 - Plugin
 - Lifecycle structure
 - `configure()` : setup parameters and get pointer
 - `activate()` : fill DStar costmap
 - `cleanup()` : destroy allocate memory
 - `createPlan()` : return a path



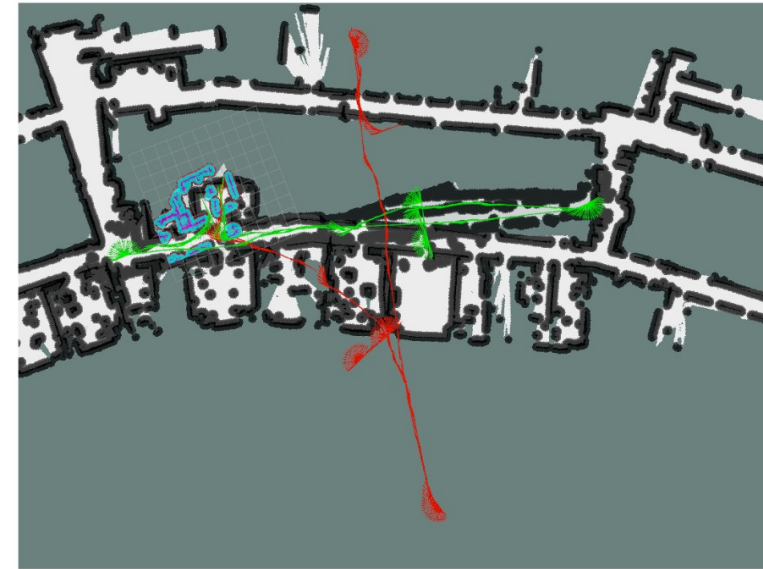
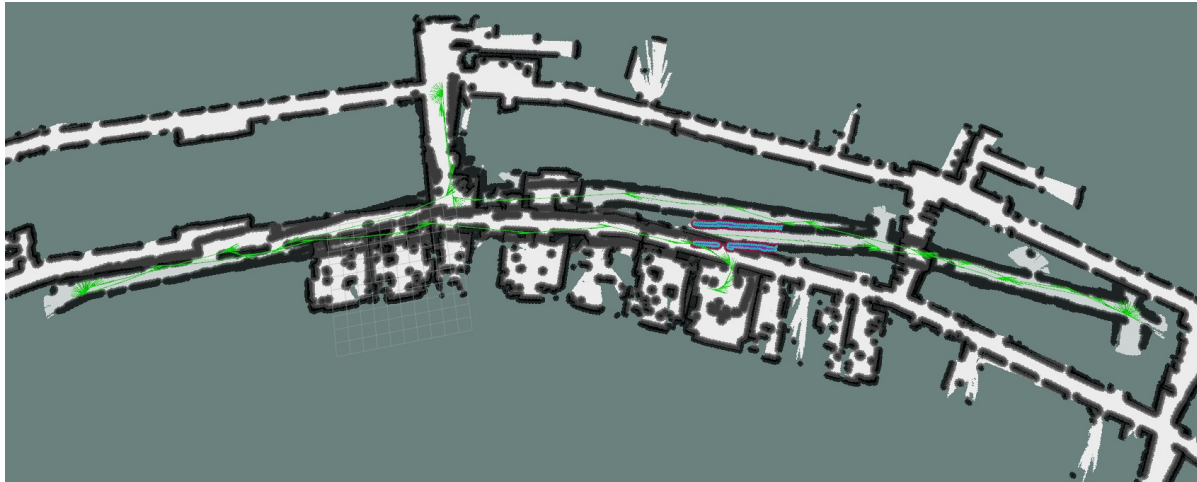
- Results:

Start->End	ROS 2 D-start (s)	ROS 2 NavfnPlanner (s)
office -> storage room	1.00	0.020
office -> entrance	5.47	0.150
entrance -> CAD designers	5.36	0.130



robot_localization

- Choosing sensor to fuse
- Calibrating the process noise covariance matrix Q of the EKF



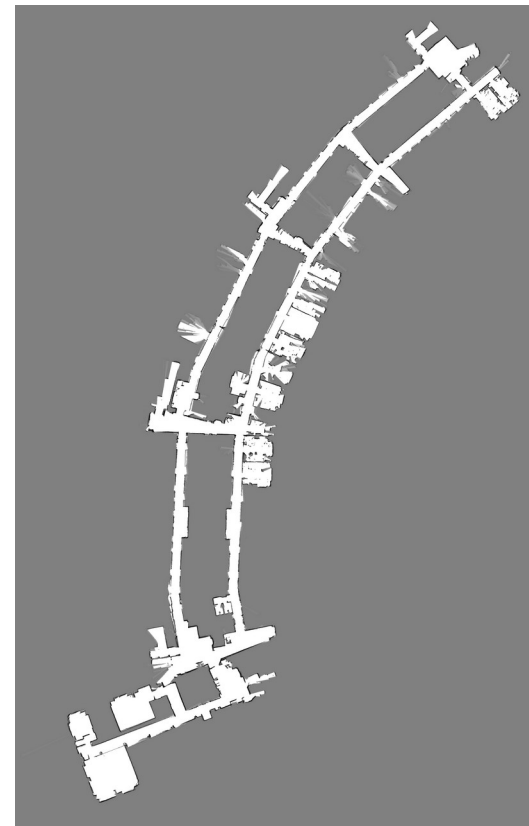
Global localization

We improve but still errors

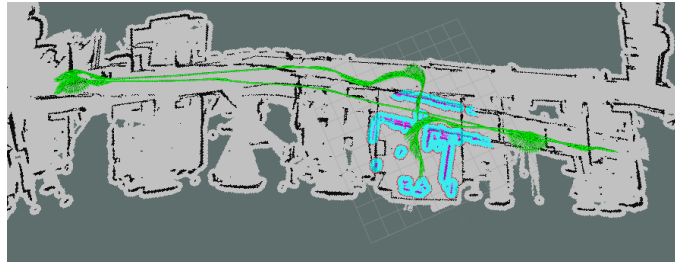


correction using a map

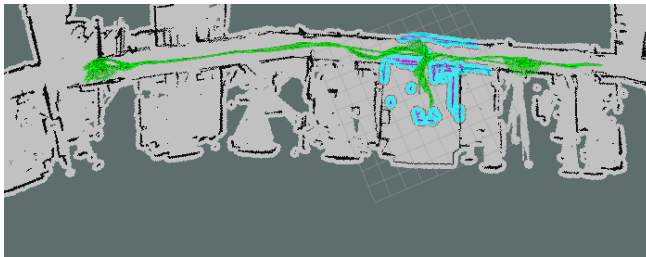
- **SLAM** or only **localization**?
 - Localization:
 - Cartographer
 - AMCL
 - Tuning:
 - α parameters :
Expected **process noise** in **odometry's rotation/translation estimate** from **rotation/translation**



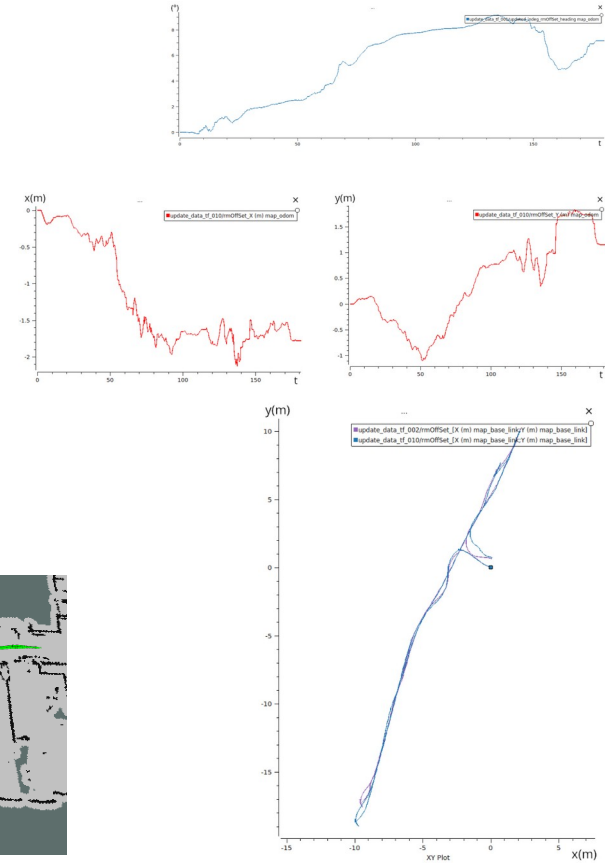
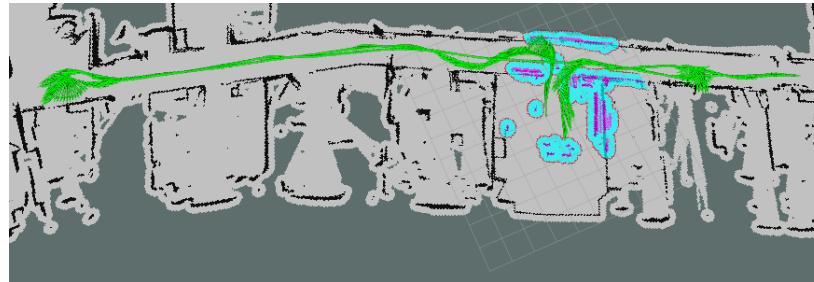
AMCL calibration Mini

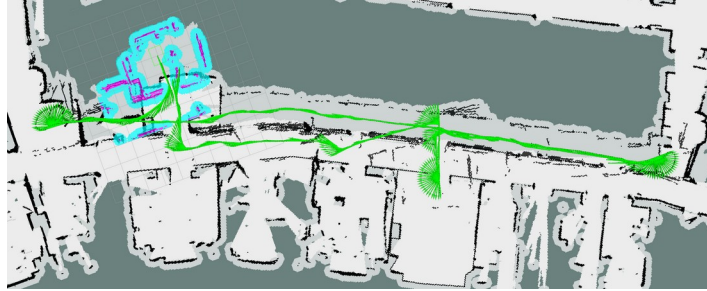


Correcting rotation

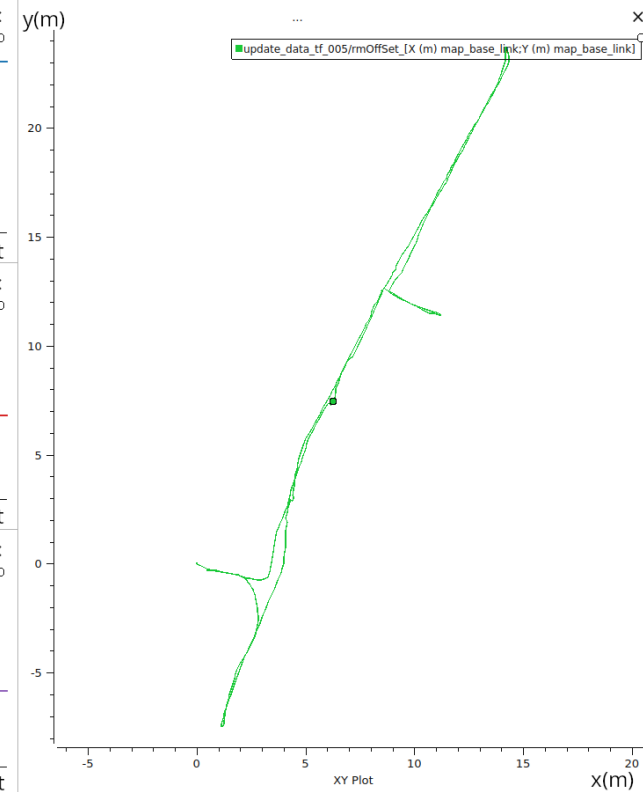
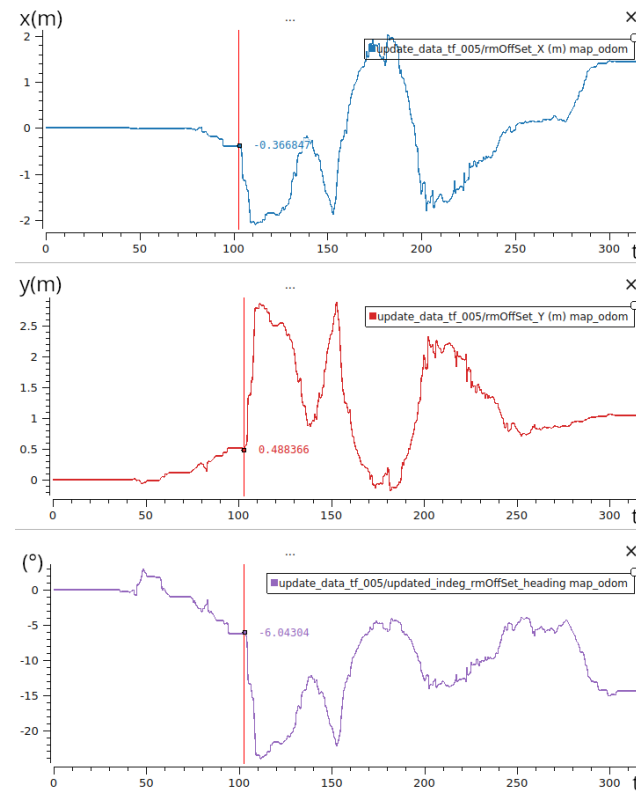


Correcting translation



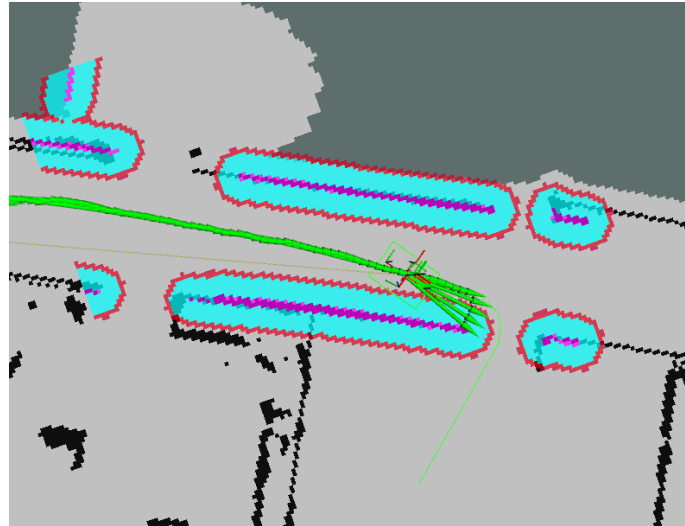
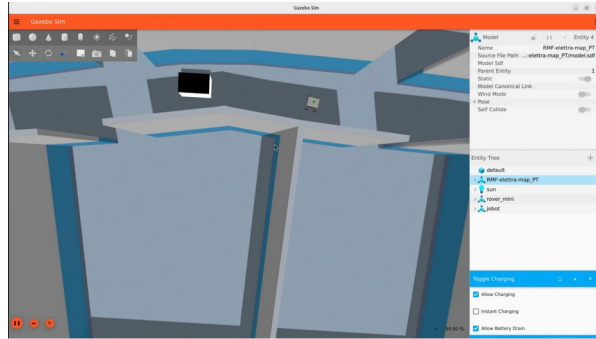


$\alpha1$ (-)	$\alpha2$ (-)	$\alpha3$ (-)	$\alpha4$ (-)	distance (m)	data name
0.0	0.0	0.0	0.0	1.63	-
0.001	0.0001	0.0001	0.0001	1.58	000
0.01	0.0001	0.0001	0.0001	2.16	001
0.01	0.001	0.0001	0.0001	0.71	002
0.01	0.001	0.001	0.0001	0.87	003
0.01	0.001	0.001	0.0001	0.74	004
0.01	0.001	0.001	0.001	0.59	005



Discovery in simulation

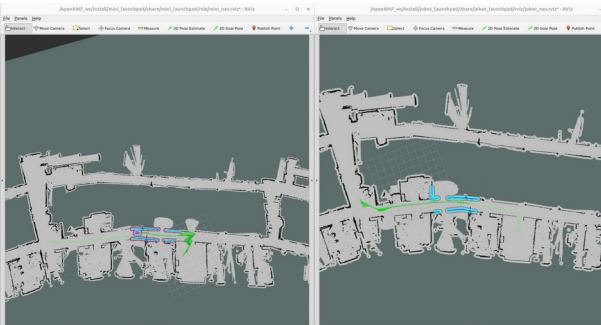
Error with the controller



DWB (Dynamic Window Approach) Controller

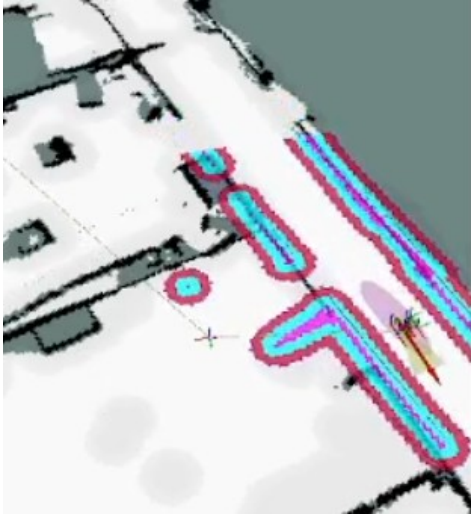


MPPI (Model Predictive Path Integral) Controller



Discoveries in reality

- Miss alignment from local costmap and global costmap

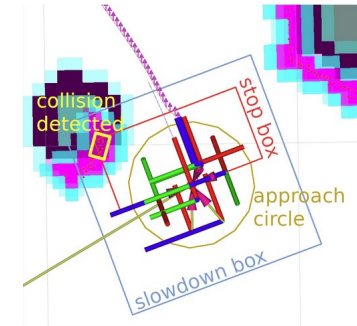
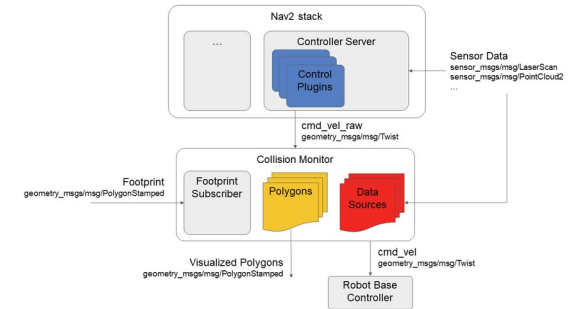


System fail to generate a path to enter the room

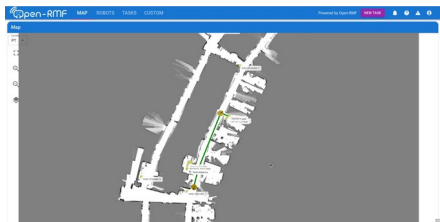
- Obstacles not visible



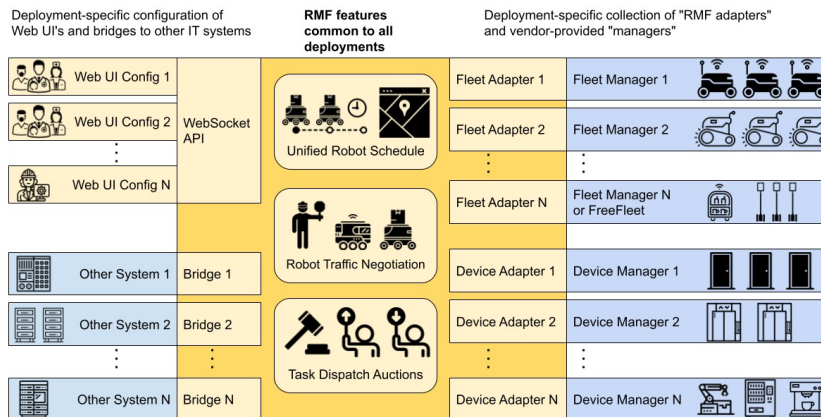
- Improve safety



Build a system manager for the fleet Mini and for the fleet Jobot



rmf_web



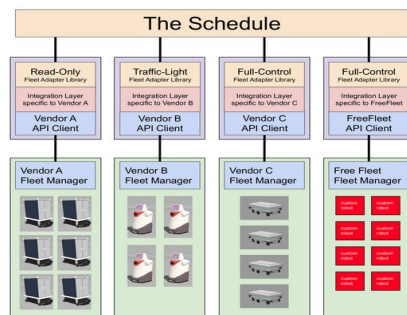
rmf_traffic_schedule

rmf_task_dispatcher

Vendor-neutral, open-source software

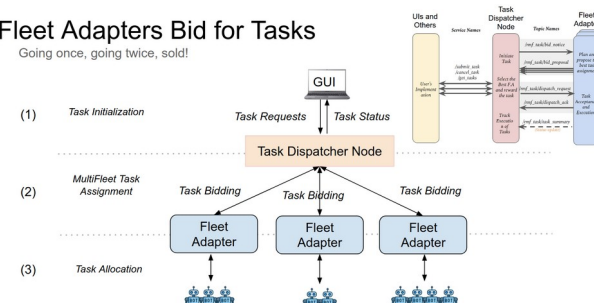
Fleet Adapters customized for each vendor

Vendor-supplied proprietary hardware and software systems



Fleet Adapters Bid for Tasks

Going once, going twice, sold!

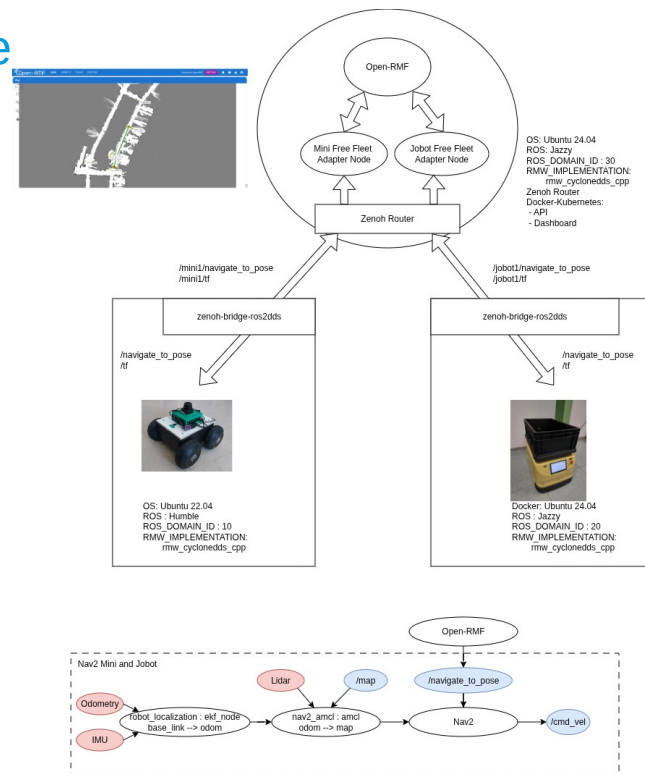
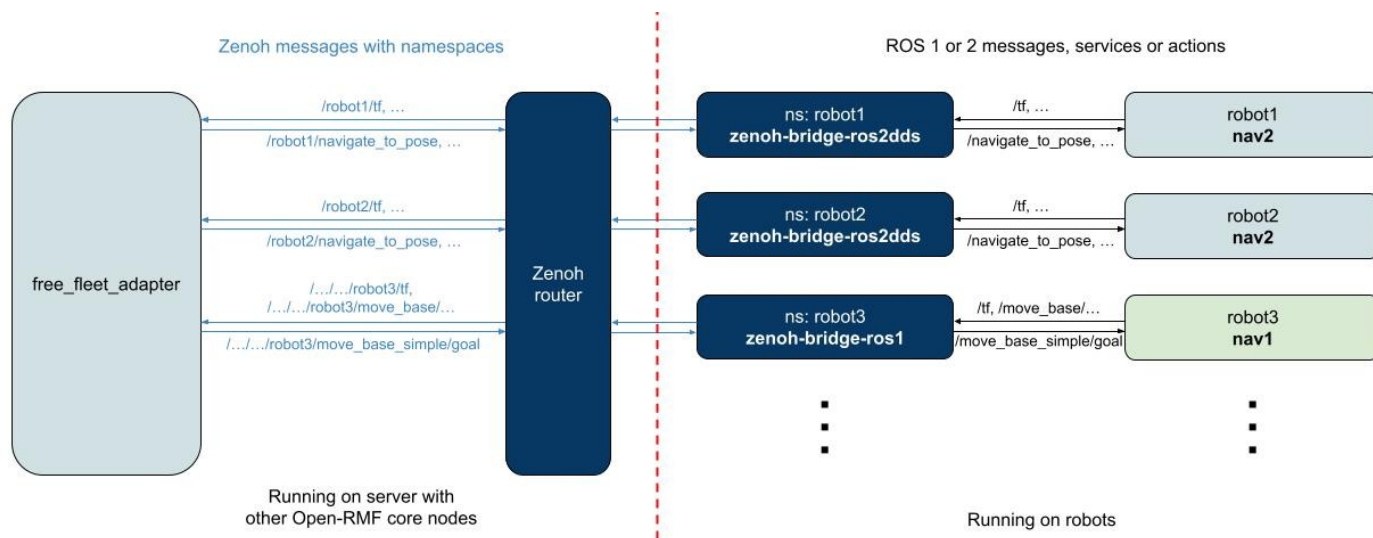


rmf_traffic_editor

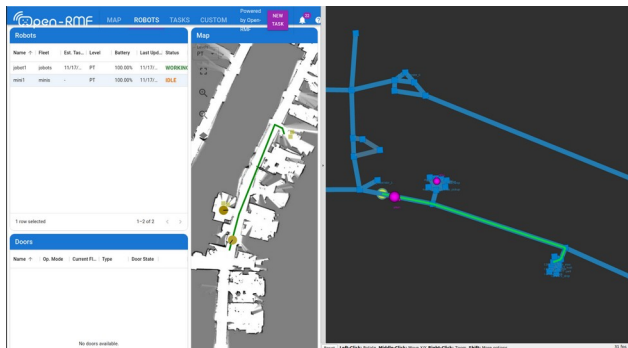


Free Fleet

a ROS 2 dedicated Fleet adapter based on fleet_adapter_template



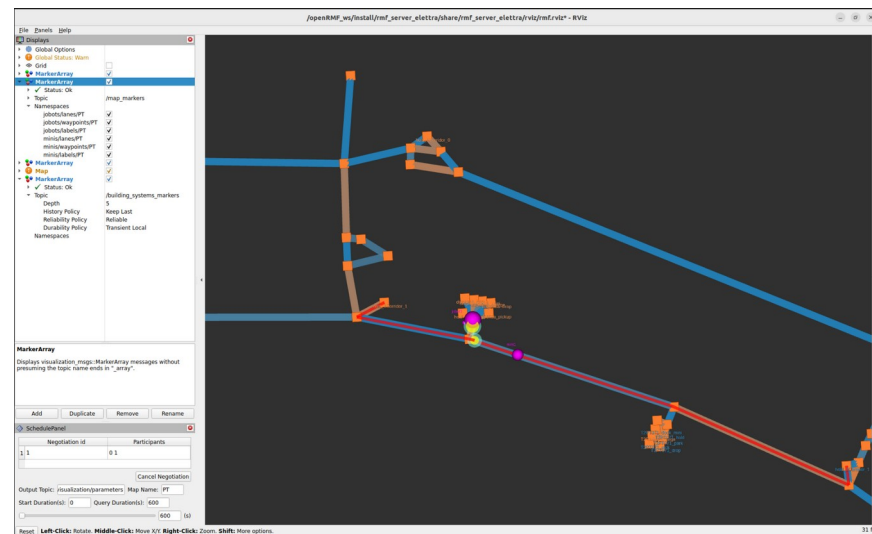
Open-RMF results



Patrol task without negotiation



Mini waiting the Jobot



Patrol task with negotiation

Thanks for the attention!

All the data is available at :

<http://dx.doi.org/10.1201/9780429489105>

While the code:

https://gitlab.elettra.eu/open-rmf-tesi/tesi_mag.git